Memorial Hermann liver-directed therapy evidence matrix

ABCDE reconciliation · 23 October 2026 · Evidence and modality selection; institutional ratification pending

Tumour	Thermal ablation	SBRT / EBRT	Y-90 (TARE)	TACE / TAE	HAI / HAIC	Histotripsy
HCC	**First-line curative ≤3 cm; ≡ resection** **Curative intent** ≤3 cm, ≤3 nodules, Child-Pugh A (BCLC 0–A); preferred ≤2 cm • **SURF RCT** RFS/OS ≡ surgery · PMID 35949295 • **SURF final** 5-y OS 74.6 vs 70.4%, HR 0.96 · PMID 40554738 • **Chen RCT** OS ≡ resection · PMID 16495695 • **Ng RCT** 10-y OS NS · PMID 29091283 • **Livraghi** ≤2 cm sustained CR · PMID 18008357 • **MWA vs RFA RCT** ≤4 cm, 2-y LTP 6 vs 12%, NS · PMID 29503247 **Curative intent** Recurrent HCC after resection; favour repeat resection if >3 cm or AFP >200 • **Xia RCT** RFA ≈ repeat hepatectomy; cx 7.3 vs 22.4% · PMID 31774468 **Curative-intent alt.** 3–7 cm: RFA + TACE • **Peng RCT** OS HR 0.53 vs RFA alone · PMID 23269991 **Bridge / adjunct** Bridge to LT • **Sapisochin ITT** dropout ≈ TACE/SBRT · PMID 28257902	**Curative-intent alternative when ablation unsuitable** **Curative-intent alt.** Single ≤5 cm (incl. recurrent), not ablatable — dome, perivascular, subcapsular, ≥2 cm • **Xi RCT vs RFA** 2-y local PFS 92.7 vs 75.8%; OS ≡ · PMID 39693584 • **Wahl** better FFLP than RFA if ≥2 cm · PMID 26628466 **Bridge / adjunct** Bridge to LT • **Sapisochin ITT** dropout 16.7% ≈ TACE/RFA · PMID 28257902 **Bridge / adjunct** Perioperative: neoadjuvant RT for resectable HCC with PVTT; adjuvant RT after narrow-margin hepatectomy • **Wei RCT** neoadjuvant: HCC death HR 0.35 vs surgery alone · PMID 31283409 • **Bai RCT** adjuvant: intrahepatic recurrence 20.8 vs 52.1% vs TACE · PMID 38488934 **Disease control / palliation** Incomplete TACE; macrovascular invasion with systemic Rx • **Comito ph III** LC ↑ vs repeat TACE · PMID 36421345 • **TRENDY ph II** TTP NS; LC ↑ vs DEB-TACE (post hoc) · PMID 37037359 • **RTOG 1112** +sorafenib OS 15.8 vs 12.3 mo · PMID 39699905 • **START RCT** TACE+EBRT > sorafenib (MVI) · PMID 29543938 **Disease control / palliation** Painful hepatic tumour (HCC or metastases): single-fraction RT • **CCTG HE1** pain response 67 vs 22% vs BSC · PMID 39245060	**Segmentectomy = curative-intent; lobar = bridge or palliative** **Curative-intent alt.** Solitary ≤8 cm, unresectable / not ablatable; ablative segmental dosimetry • **LEGACY** ORR 88%, 3-y OS 86.6% · PMID 33739462 • **RASER** ≤3 cm, sustained CR 90% · PMID 35617978 **Bridge / adjunct** Bridge / downstage to LT or resection; radiation lobectomy for FLR • **PREMIERE RCT** TTP >26 vs 6.8 mo vs cTACE · PMID 27575820 • **TRACE RCT** OS 30.2 vs 15.6 mo vs DEB-TACE · PMID 35258371 • **MERITS-LT** Y90 ≈ TACE for downstaging · PMID 34331914 • **Vouche** radiation lobectomy: FLR +45% at 9 mo · PMID 23811303 **Disease control / palliation** Advanced / PVTT: no OS gain vs sorafenib; dosimetry matters • **SARAH** OS NS · PMID 29107679 • **SIRveNIB** OS NS · PMID 29498924 • **SORAMIC** OS NS · PMID 31421157 • **DOSISPHERE-01** ORR 71 vs 36% · PMID 33166497	**Standard for BCLC-B; main bridging / downstaging tool** **Bridge / adjunct** Ruptured HCC: TAE for haemostasis, then staged hepatectomy • **Meta-analysis (18 studies)** lower in-hospital mortality, better 1-/3-y OS vs emergency hepatectomy · PMID 37328589 **Bridge / adjunct** Adjuvant after resection, intermediate/high recurrence risk • **Wang RCT (HBV)** RFS ↑; 3-y OS 85.2 vs 77.4% · PMID 29420221 **Bridge / adjunct** Bridge / downstage to LT (UNOS-DS) • **XXL RCT** LT after downstaging ↑ OS · PMID 32615109 • **MERITS-LT** >80% downstaged; 2-y post-LT OS 95% · PMID 34331914 **Disease control / palliation** BCLC-B multinodular, Child-Pugh A–B7, no MVI; ± systemic • **Llovet RCT** OS HR 0.47 vs supportive · PMID 12049862 • **Lo RCT** OS ↑ vs supportive · PMID 11981766 • **EMERALD-1** +durva/bev PFS 15.0 vs 8.2 mo · PMID 39798579 • **LEAP-012** +len/pembro PFS 14.6 vs 10.0 mo · PMID 39798578	**Large / advanced HCC — mostly Asian RCTs** **Disease control / palliation** Large unresectable, high burden, PVTT (Vp3–4) • **FOHAIC-1** OS 13.9 vs 8.2 mo vs sorafenib · PMID 34905388 • **He RCT** Sora+HAIC OS 13.4 vs 7.1 mo · PMID 31070690 • **Li ph III** HAIC > TACE, OS 23.1 vs 16.1 mo · PMID 34648352 **Bridge / adjunct** Conversion in responders • **FOHAIC-1** 12% downstaged to surgery/ablation · PMID 34905388 **Bridge / adjunct** Adjuvant HAIC-FOLFOX after resection for MVI • **Li ph III** DFS 20.3 vs 10.0 mo; OS NS · PMID 36525610	**FDA-cleared 2023; no long-term oncologic data** **Investigational** ≤3 tumours <3 cm with acoustic window; trial / registry • **#HOPE4LIVER** technical success 95%, major cx 7% · PMID 39225612 • **#HOPE4LIVER 1-y** 1-y LC 63–90%; 1-y OS 73% (HCC) · PMID 40201962
iCCA	**Small primary or recurrent iCCA when resection not chosen** **Curative-intent alt.** ≤3 cm, ≤3 nodules; unresectable / unfit, or intrahepatic recurrence; margin ≥5 mm • **MALRIC (PSM)** MWA vs resection 5-y OS 44.8 vs 40.4% · PMID 38261915 • **MALRIC recurrent** MWA ≈ repeat resection OS · PMID 41528222 • **Meta-analysis** worse if >3 cm or multiple · PMID 34347159	**Ablative-dose RT for localized unresectable iCCA** **Curative-intent alt.** Liver-confined, unresectable, no nodal/EHD; BED >80 Gy • **Tao** 3-y OS 73% (BED >80.5) vs 38% · PMID 26503201 • **Hong ph II (proton)** 2-y LC 94%, OS 46.5% · PMID 26668346	**Add to 1st-line gem/cis; minority downstaged** **Disease control / palliation** Unresectable, liver-confined • **MISPHEC ph II** +gem/cis, mOS 22 mo · PMID 31670746 **Bridge / adjunct** Downstaging to resection • **MISPHEC ph II** 22% downstaged, 20% R0 · PMID 31670746 **Bridge / adjunct** Radiation lobectomy to grow FLR before resection • **Vouche (mixed series)** few non-HCC cases · PMID 23811303	**Palliative option; low-level evidence** **Disease control / palliation** Unresectable liver-dominant, with systemic therapy • **Kiefer 2-centre** mOS 15 mo; 28 vs 16 mo with systemic · PMID 21425151	**Expert-centre control / conversion; no comparative superiority** **Disease control / palliation** Unresectable liver-confined disease ± selected resectable regional nodes; systemic-treatment plan required Pump FUDR evidence is distinct from pCCA catheter GemCis plus I-125 stents. • **Cercek ph II** 6-mo DCR 84%, mOS 25 mo · PMID 31670750 • **Konstantinidis** HAI+sys vs sys OS 30.8 vs 18.4 mo · PMID 26695839 • **PUMP-2 phase II** 50 patients; median OS 22.3 months, 3-y OS 31.5%; historical control · PMID 41082695 • **Pooled four phase II trials** 142 treated; median OS 26 months, 5-y OS 15%; overlapping cohorts · PMID 42331230 **Bridge / adjunct** Conversion to resection • **Cercek ph II** 4 of 38 resected · PMID 31670750 • **PUMP-2** 5/50 (10%) resected; no randomized comparison with modern chemo-immunotherapy · PMID 41082695	**No iCCA data** **Investigational** Trial only — pivotal trial enrolled HCC and metastases, not iCCA • **#HOPE4LIVER** · PMID 39225612
pCCA / hCCA	**No established percutaneous thermal ablation role** **Investigational** Central bile ducts and adjacent vessels limit safe margins. Endobiliary RFA is a separate palliative technique. Review transplant candidacy before intervention; protect the hilar ducts and vasculature. The explicit OPTN primary-tumour biopsy prohibition must not be generalized to every procedure. • **BSG guideline** Primary hilar duct disease differs from mass-forming iCCA. · PMID 37770126	**Protocol RT before LT; selected non-LT control** **Bridge / adjunct** Curative-intent LT pathway: follow the approved neoadjuvant protocol, then operative staging and LT. OPTN 9.5.A requires neoadjuvant therapy, not a specific RT technique or Mayo regimen. Chemotherapy-only selection remains a protocol-level decision. • **Darwish Murad, 12 centres** 287 enrolled; 5-year intention-to-treat overall survival 53% from start of therapy; 5-year post-transplant recurrence-free survival 65%. No isolated RT effect. · PMID 22504095 • **Welling pilot, Liver Transpl 2014** 17 evaluated; 12 RT-treated (11 SBRT), 9 listed, 6 LT. SBRT 50–60 Gy/3–5 fx; 1-y post-LT OS 83%. · PMID 24115315 • **Soliman, Transplant Direct 2025** 26 LT recipients: 12 GemCis-only vs 14 RT-containing. No significant OS/RFS difference; no equivalence or ITT benefit established. · PMID 39936132 • **Hoogwater, BJS Open 2023** 49 LT recipients, 27 neoadjuvant chemoradiation vs 22 no neoadjuvant therapy. Lower adjusted recurrence hazard, more vascular complications; no proven OS benefit or equivalence. · PMID 37032423 **Disease control / palliation** Unresectable pCCA outside LT: NCCN lists chemoradiation with or without sequential chemotherapy as an option, without requiring prior systemic control (v1.2026, original PDF p23, EXTRA-1). BSG supports selected SBRT/PBT after systemic therapy. Individualize in MDT; trial preferred. No proven routine survival advantage; drainage and organ-at-risk constraints govern feasibility. • **STRONG phase I** 6 patients, 60 Gy/15 fractions; 1-y LC 80% from RT. Grade 3 cholangitis in 5/6; no randomized survival benefit. No dose-limiting toxicity used grade ≥4 hepatobiliary or grade ≥3 gastrointestinal criteria; grade 3 cholangitis was not a DLT. · PMID 34439146 • **ABC-07 randomized phase II** ASCO 2024 abstract 4006. n=69, 84% perihilar. PFS 8.6 vs 9.0 months, HR 1.00; no significant OS benefit. DOI 10.1200/JCO.2024.42.16_suppl.4006 • **BSG recommendation 43** Supports consideration of SBRT/PBT after systemic therapy; benefit evidence limited. · PMID 37770126 • **STRONG II protocol** Phase II target n=30; protocol, not outcome evidence. · PMID 40669914 • **ABC-07 RTQA, Eaton 2026** 19/41 plans deviated from protocol objectives, largely organ-at-risk constraints. Does not establish efficacy. · PMID 41512568 • **STRONG II benchmark QA 2026** Planning variability study, not patient survival or local-control results. · PMID 42744077 **Bridge / adjunct** Post-resection: NCCN lists chemoradiation for R1 or node-positive eCCA and as a nonpreferred option for R0/node-negative disease (v1.2026, original PDF p24, EXTRA-2). Systemic therapy and clinical trial are preferred. BSG specifically supports R1 discussion. No pCCA-specific randomized proof of RT survival benefit. • **BSG recommendation 42** Shared decision about potential local benefit and toxicity; no pCCA-specific randomized proof. · PMID 37770126	**No established routine role for primary pCCA** **Investigational** Do not extrapolate iCCA radioembolization series to infiltrative hilar duct tumors. Any liver-parenchymal target requires individualized MDT review and biliary safety assessment. Review transplant candidacy before intervention; protect the hilar ducts and vasculature. The explicit OPTN primary-tumour biopsy prohibition must not be generalized to every procedure. • **BSG guideline** Primary hilar duct disease differs from mass-forming iCCA. · PMID 37770126	**No established routine role for primary pCCA** **Investigational** No curative or standard sequencing evidence for the primary hilar tumor. Obstruction, biliary instrumentation and infection heighten concern. Review transplant candidacy before intervention; protect the hilar ducts and vasculature. The explicit OPTN primary-tumour biopsy prohibition must not be generalized to every procedure. • **BSG guideline** Primary hilar duct disease differs from mass-forming iCCA. · PMID 37770126	**HAIC evidence is exploratory and regimen-specific** **Investigational** Chinese trial tested radiation-emitting vs conventional stents, BOTH with arterial gemcitabine/cisplatin. It did not test HAIC versus systemic therapy or pump FUDR. • **Chen randomized trial** 126 patients; OS 10.2 vs 6.7 months. Effect is the stent comparison within HAIC, not proof of HAI benefit. · PMID 39311927	**No established primary hilar treatment role** **Investigational** No pCCA-specific curative, survival or transplant-bridge evidence. Central hilar anatomy requires research-protocol assessment. Review transplant candidacy before intervention; protect the hilar ducts and vasculature. The explicit OPTN primary-tumour biopsy prohibition must not be generalized to every procedure. • **BSG guideline** Primary hilar duct disease differs from mass-forming iCCA. · PMID 37770126
CRLM	**Curative; ≡ resection for ≤3 cm** **Curative intent** ≤3 cm (≤10 lesions), resectable; margin >5 mm • **COLLISION RCT** OS non-inferior (HR 1.05); AEs 19 vs 46% · PMID 39848272 • **Shady** <3 cm & margin >5 mm drive LC · PMID 26267832 **Curative-intent alt.** Unresectable liver-only, with systemic therapy • **CLOCC RCT** 8-y OS 35.9 vs 8.9% · PMID 28376151	**Curative-intent when not resectable / ablatable** **Curative-intent alt.** 1–3 lesions, ≤6 cm, not resectable or ablatable • **Rusthoven ph I/II** 2-y LC 92% (100% if ≤3 cm) · PMID 19255321 • **SABR-COMET** mixed oligomets incl. CRC · PMID 30982687 • **SABR-COMET LT** 5-y OS 42.3 vs 17.7% · PMID 32484754	**Salvage for chemo-refractory liver-dominant disease** **Disease control / palliation** Chemo-refractory or 2nd-line, liver-only / dominant • **EPOCH ph III** PFS 8.0 vs 7.2, hPFS 9.1 vs 7.2 mo; OS NS · PMID 34541864 • **Hendlisz ph III** TTLP 5.5 vs 2.1 mo · PMID 20567019 **Bridge / adjunct** Radiation lobectomy to grow FLR before resection • **Vouche (mixed series)** few non-HCC cases · PMID 23811303 **Investigational** Not for 1st-line • **FOXFIRE/SIRFLOX/-Global** no OS benefit · PMID 28781171	**Limited; DEBIRI only as salvage / in trials** **Disease control / palliation** Chemo-refractory liver-dominant (salvage) • **Fiorentini RCT (n=74)** DEBIRI vs FOLFIRI OS 22 vs 15 mo · PMID 22493375	**Early conversion review; selective adjuvant use** **Bridge / adjunct** Adjuvant after resection: selective expert-centre discussion, not routine low-risk use Pump FUDR evidence is distinct from pCCA catheter GemCis plus I-125 stents. • **Kemeny NEJM RCT** 2-y hepatic RFS 90 vs 60% · PMID 10615075 • **Groot Koerkamp** PSM OS HR 0.67 · PMID 28426374 • **Dutch PUMP, ASCO 2026 LBA3506** PFS HR 0.88 (0.62–1.25), P=.48; no adjuvant systemic therapy; OS immature. DOI 10.1200/JCO.2026.44.17_suppl.LBA3506 **Bridge / adjunct** Conversion of unresectable liver-only disease • **Kemeny ph I** 47% converted to resection · PMID 19470932 • **D’Angelica phase II** 49 patients; 47% resected, 76% response. Distinct from the 49-patient phase I trial. · PMID 24646562 **Disease control / palliation** Unresectable liver-only • **CALGB 9481** OS 24.4 vs 20 mo vs systemic · PMID 16505413 **Disease control / palliation** Salvage liver-only disease with systemic plan • **Cercek retrospective** 57 liver-only patients: response 33%, median OS 20 months; selected cohort · PMID 27566258 **Investigational** HAI-to-LT planning: parallel transplant review; fixed bridge schedule investigational • **Hill technical series** 7 transplants: 5 for biliary injury, 2 residual tumour; difficult arterial reconstruction · PMID 37210451	**No CRLM-specific oncologic data** **Investigational** Trial / registry only • **#HOPE4LIVER** 59% had metastases (mixed) · PMID 39225612 • **#HOPE4LIVER 1-y** 1-y OS 48.6% (metastases) · PMID 40201962
NELM	**Limited small lesions; adjunct to resection / debulking** **Curative-intent alt.** Few small lesions, alone or combined with resection • **Mazzaglia (n=63)** LTP 6.3%; mOS 3.9 y post-RFA · PMID 17629995 • **Akyildiz** OS 6 y post-RFA · PMID 21134563 **Bridge / adjunct** Cytoreduction / hormonal symptom control • **Mazzaglia** symptom relief in most · PMID 17629995	**No NET-specific series** **Investigational** Consider for 1–3 lesions not resectable / ablatable (extrapolated) • **SABR-COMET (mixed)** no NET-specific data · PMID 30982687	**Bulky liver-dominant disease; late hepatotoxicity** **Disease control / palliation** Unresectable, progressive or symptomatic, liver-dominant • **Kennedy (n=148)** PR 60.5%, mOS 70 mo · PMID 18525307 **Caution** Bilobar → cirrhosis-like change / portal HTN; plan sequence with PRRT • **Tomozawa** 15 of 52 with >1-y follow-up · PMID 29724520	**Standard embolotherapy; avoid DEB-TACE** **Disease control / palliation** Progressive or symptomatic liver-dominant disease • **Gupta (n=123)** carcinoid > islet-cell response · PMID 16134179 • **RETNET RCT** TAE ≈ cTACE, hPFS 18 vs 13 mo, NS* · PMID 30016989 • **Bhagat ph II** DEB-TACE: biloma 54% · PMID 22722717	**No established role** **Investigational** No dedicated data	**No NET-specific data** **Investigational** Trial only • **#HOPE4LIVER** mixed metastases · PMID 39225612
Contraindications / feasibility	• Size: >3 cm (RFA) or >5 cm (MWA) for curative intent; >3–5 lesions (relative) • Not visible on US/CT or no safe needle path • <1 cm from hilar / central bile duct (stricture) • Abutting bowel, stomach, GB, diaphragm or heart without hydro-/pneumodissection • Heat sink: abutting vessel ≥3 mm (RFA > MWA) • Bilio-enteric anastomosis, sphincterotomy or stent (abscess; relative) • Platelets <50k / INR >1.5 uncorrectable; Child-Pugh C; refractory ascites CIRSE SoP · 32382856	• Inadequate uninvolved liver (≥700 mL must be spared); Child-Pugh ≥B8 relative, C absolute • Stomach / duodenum / bowel abutting target (dose limits) • Overlapping prior hepatic RT • Cannot manage motion (fiducials, breath-hold) or lie flat • Very large or multifocal disease (>6–8 cm; relative) HyTEC liver SBRT · 29482870	• Lung shunt: lung dose >30 Gy single / >50 Gy cumulative • Uncorrectable GI / extrahepatic deposition on MAA • Bilirubin >2 mg/dL (unless segmental); Child-Pugh C; ECOG ≥2 • Insufficient hepatic reserve; extensive tumour replacement with poor function • Prior biliary intervention or prior hepatic RT (relative) • Note: PVTT is NOT a contraindication Y-90 recs 2022 · 36114872	• Main PV occlusion / hepatofugal flow without collaterals (selective TACE may still be possible) • Decompensated cirrhosis: Child-Pugh C (≥B8 relative), bilirubin >3 mg/dL, refractory ascites, HE • Tumour burden >50% of liver • Bilio-enteric anastomosis, stent or sphincterotomy (abscess) • CrCl <30, uncorrectable coagulopathy, large AV shunt • DEB-TACE in NET (biliary injury) CIRSE SoP · 34694454	• Uncorrectable extrahepatic perfusion or inadequate hepatic perfusion • Unsuitable arterial anatomy or impaired portal inflow; significant portal hypertension or poor hepatic reserve • Unrelieved biliary obstruction / infection; unsuitable for pump surgery • No reliable refill, laboratory surveillance or rescue service • Prior liver radiation requires review before sequencing; retain NCCN CRLM restriction • Avoid concurrent bevacizumab with FUDR. Disease extent and transplant eligibility remain tumour-specific. Surgical toolkit / programme requirements / biliary toxicity · 36346892	• No acoustic window: rib / lung shadow or bowel gas in beam path • Target not visible on diagnostic US or beyond transducer focal depth • Unable to tolerate general anaesthesia (needed for respiratory control) • Beyond trial limits: >3 tumours or >3 cm • Lung or bowel immediately adjacent to the treatment zone #HOPE4LIVER · 39225612

ABCDE review, 23 October 2026: tiers describe clinical intent, not guaranteed cure, a formal evidence grade, or approval. Read each endpoint and time origin. pCCA RT may support protocol LT or selected local control; RT is not specifically mandated by OPTN. Clinical benefit, allocation eligibility and institutional approval are separate. Other tumour selection criteria were not expanded by this pCCA review.